Machine Learning for Time Series project

Henri Fabre henri.fabre@ensae.fr
Jean-Lou Valeau jeanlou.valeau@ensae.fr

1 Introduction

Changepoint detection is a task aiming to detect the moments at which a time series change patterns, for instance when the underlying distribution changes. The article we study, [1], proposes a novel method to detect changepoints while not overfitting on outliers. This method is of particular interest for noisy data. [1] introduces a penalized loss to account for outliers while detecting a changepoint. For efficient computing, they introduce a dynamic programming algorithm enabling an efficient solving of the cost minimization.

We use the method developed by [1] to detect changepoints in several macroeconomic time series. Changepoint detections is particularly valuable for macroeconomists as it can automatically detect a structural change in a series of interests. For instance, detecting a change in inflation expectations could infer long-term changes in inflation patterns and change the rule with which the central bank sets its rates. Hence, changepoint detection is a dynamic sector of research in economics (see, eg [3]). Having ourselves an interest in economics from our studies and professional experiences, we decided to apply our methods to several macroeconomic time series, namely the 5 and 10-year inflation expectation, the 10-year inflation break-even and GDP time series.

We conducted the project as follows. Jean-Lou coded the dynamic programming algorithms and applied them to synthetic data to replicate the results of the papers. He also did the exploratory phase of our applications, and coded a sensitivity analysis of the number of changepoints as a function of beta, the penalty parameter. Henri coded a gridsearch as well as an elbow method in order to systematically tune the parameters, and did an online analysis. Jean-Lou recoded all the algorithms himself, since there is no available rfpop package in Python. We used existing experiments on synthetic data but we did several new experiments :

- We tested the influence of the β parameter on the number of changepoints.
- We applied the method to the automatic detection of changepoints in inflation and GDP-related macroeconomic time series.
- We added a gridsearch on the parameters of interest, as well as an elbow method to find the optimal number of changepoints.

2 Method

The method developed by [1] relies on two main components: a penalized cost function that is robust to outliers, and an efficient dynamic programming algorithm to minimize this cost. We summarize here the key mathematical aspects.

2.1 Penalized Cost Framework

Consider a time series $y = (y_1, \ldots, y_n)$ with k changepoints at locations $\tau = (\tau_1, \ldots, \tau_k)$, splitting the data into $k+1$ segments. For each segment, we assume a location parameter θ (e.g., the mean). The cost of fitting a segment $y_{s:t}$ is defined as:

$$C(y_{s:t}) = \min_{\theta} \sum_{i=s}^{t} \gamma(y_i; \theta), \tag{1}$$

where $\gamma(y; \theta)$ is a loss function measuring the discrepancy between observation y and parameter θ. The penalized cost for a segmentation is then:

$$Q(y_{1:n}; \tau_{1:k}) = \sum_{i=0}^{k} \left[C(y_{\tau_i+1:\tau_{i+1}}) + \beta \right], \tag{2}$$

where $\beta > 0$ penalizes the addition of changepoints. The optimal segmentation is obtained by minimizing Q over all possible k and $\tau_{1:k}$.

2.2 Robust Loss Functions

The standard L2 loss $\gamma(y; \theta) = (y - \theta)^2$ is highly sensitive to outliers. A common alternative is the Huber loss, which grows linearly rather than quadratically for large errors:

$$\gamma(y; \theta) = \begin{cases} (y - \theta)^2 & \text{if } |y - \theta| \leq K, \\ 2K|y - \theta| - K^2 & \text{otherwise.} \end{cases} \tag{3}$$

where K is a threshold parameter. Another loss discussed by the paper is the biweight loss:

$$\gamma(y; \theta) = \begin{cases} (y - \theta)^2 & \text{if } |y - \theta| < K, \\ K^2 & \text{otherwise.} \end{cases} \tag{4}$$

This loss behaves like the L2 for observations close to θ, but caps the contribution of outliers at K^2. A crucial theoretical result (Theorem 2.2 in [1]) states that if $\gamma(y; \theta) \leq K$, all inferred segments have length greater than β/K. This minimum segment length provides robustness: isolated outliers cannot be placed in their own segment, preventing spurious changepoint detection. However, this also implies that consecutive outliers exceeding this minimum length may still be misidentified as true changepoints.

2.3 The R-FPOP Algorithm

Minimizing the penalized cost over all possible segmentations is computationally challenging, as the number of possible segmentations grows exponentially with the length n of the time series. The authors propose R-FPOP (Robust Functional Pruning Optimal Partitioning), an extension of the FPOP algorithm.

The algorithm relies on a dynamic programming recursion. Define $Q_t = \min_{\tau} Q(y_{1:t}; \tau)$ as the minimum penalized cost for segmenting data up to time t. Introducing $Q_t(\theta)$, the minimum cost conditional on the last segment having parameter θ, we have $Q_t = \min_{\theta} Q_t(\theta)$ and:

$$Q_t(\theta) = \min\{Q_{t-1}(\theta), Q_{t-1} + \beta\} + \gamma(y_t; \theta). \tag{5}$$

The key observation is that for piecewise quadratic loss functions (which includes the biweight loss), $Q_t(\theta)$ remains piecewise quadratic in θ. The algorithm maintains a representation of $Q_t(\theta)$ as a collection of intervals with associated quadratic functions. At each step t, three operations are performed:

1. Compute the pointwise minimum between $Q_{t-1}(\theta)$ and the constant $Q_{t-1} + \beta$;
2. Merge neighboring intervals that share the same functional form;
3. Add the loss $\gamma(y_t; \theta)$ for the new observation.

For convex loss functions, R-FPOP yields $O(n^2)$ time complexity and $O(n)$ space complexity (Corollary 4.2 in [1]). For the biweight loss, which is non-convex, the time complexity is $O(n^3)$ and space complexity is $O(n^2)$ (Corollary 4.4 in [1]).

Implementation details: In practice, the authors set $K = 3\hat{\sigma}$, where $\hat{\sigma}$ is the median absolute deviation of the differenced series. This choice ensures that observations within three standard deviations contribute their full squared residual, while more extreme values are treated as potential outliers. For the penalty, we use $\beta = 2\hat{\sigma}^2 \log(n)\mathbb{E}[\phi(Z)^2]$, where ϕ is the gradient of the loss and $Z \sim \mathcal{N}(0,1)$, corresponding to a modified BIC criterion. However, as we will see in the results, these recommendations are not appropriate for very noisy series. We will need to find another way of inferring parameters.

3 Data

We first implemented experiments on synthetic data corresponding to a simulation benchmark proposed in the seminal paper [2] and used in [1], then we conducted new experiments on macroeconomic time series sourced from the FRED dataset.

3.1 Synthetic data

Each of the six time series corresponds to a set of different segments (the junction bewteen two segments corresponding to a changepoint), to which we add noise drawn from a $t-$distribution with 5 degrees of freedom and with varying levels of variance. These series vary in length, length of segments, space between changepoints, etc. The reason for using a t-distribution to inject noise in the data is that it is a heavy-tailed distribution, which makes it ideal to assess the robustness of our models to outliers. Due to the way these scenarios were generated, data is non-stationary and there are different regimes with different means: this is obvious when plotting the time series and their rolling mean, which varies greatly over time (see Figure 1 in appendix). This non-stationarity therefore justifies the use of R-FPOP to detect the different segments.

3.2 Real data

We used the T5YIE and T10YIE inflation-related series, which respectively represent the market's expected average inflation rate over the next 5 or 10 years. We also used the T5YIFR series, which measures the expected inflation over 5 years, in 5 years: it is commonly used to assess how well long-term inflation expectations are anchored, filtering out short-term shocks. These time series are especially relevant for changepoint detection in the presence of outliers as they're based on market expectations so they're very noisy and contain many points taking extreme values. Furthermore, these data contain different regimes associated with different periods in US economic

history. Before 2008, inflation expectations were stable at 2.5% as markets were confident in the Fed's ability to control inflation. In 2008, the crisis hit and the financial instrument measuring inflation was sold off massively, artificially driving inflation expectations down to 0 (which corresponds to measurement noise, not a real signal). From 2014 to 2016, there was an oil crash which brought inflation expectations down to 1.5% as inflation is driven by energy prices. Since 2021, the fiscal stimulus caused by supply chain shocks have pushed inflation expectations back up to 2.5%. These different economic regimes make the data non-stationary, as evidenced by the variability of the rolling mean and variance of the data (see Figure 2). When plotting the autocorrelation of the data, it remains large significative even after 40 lags which confirms non-stationary (see Figure 2).

We also used the JPNRGDPEXP, CLVMNACSCAB1GQDE, UKNGDP time series, which respectively represent the Real GDP time series of the Japan, Germany and the UK. These GDP time series are also ideal for applying changepoint detection methods as there are many outliers caused by exogenous economic shocks such as the covid pandemic, some natural catastrophes, wars, or financial crisis. Similarly, there are different regimes corresponding to various chapters of the economic history of these countries (periods of boom and recession) so data is not stationary, as exemplified by the rolling mean and rolling variance of the series and their significative autocorrelations (see Figure 3).

4 Results

4.1 Results on simulated data

We first tried using the values of K and β recommended in the section 5.2 of the paper but this resulted in a massive over-segmentation problem in some cases, with hundreds of change points detected. In other cases, the model resulted in under-segmentation and there were not enough changepoints detected. This means that the parameter values recommended by the article result in either a lack of penalty or excessive penalty. It therefore appears that the models presented in the article are very sensitive to parameter tuning.

To obtain changepoints that matched the actual signal, we had to trial and error and adjust β and K iteratively. After these lengthy adjustments, we managed to obtain very satisfactory results with the Huber loss (Figure 4) and the L2 loss (Figure 5). Surprisingly, it was impossible to obtain decent results with biweight loss, which contradicts the article: there was a systematic problem of over/under segmentation. The biweight loss therefore seems to be too sensitive to the values of β and K on this data, showing that this loss isn't adapted to all types of data.

4.2 Results on economic time series

As for synthetic data, running the algorithm on real data with values of β and K recommended by the paper consistently yielded over-segmentation that didn't match the expected historical structural breaks (figure 6). We therefore had to manually and incrementally adjust the penalty parameter to obtain economically meaningful results. In order to do this, we ran sensitivity analyses, which showed that the scale of beta suggested by the paper is really low compared to what should be expected for such data. We therefore opted for higher parameters for our final analyses. Higher parameters enabled us to get interesting findings (see figure 7) and to match the structural breaks mentioned in section 3.2.

Inflation expectations (T5YIFR, T10YIE). All three loss functions successfully identify the major eras of recent US monetary history: the structural break of 2014 (shift to a low-inflation regime)

and the inflationary shock of 2021 (post-Covid inflation and Ukraine war). The L2 and Huber losses (left and right plot on figure 7) also detect the 2008 Global Financial Crisis and the 2020 COVID shock as distinct regimes, which makes sense from an investor's perspective. The Biweight loss (plot in the middle on figure 7) treats these crashes as outliers rather than structural changes, providing a view that filters out market panic to focus on fundamental shifts. This interpretation aligns with a monetary policy evaluation perspective. Our online analysis shows that the online process might lack stability, as some changepoints revert and the function of the number of changepoints as a function of time is not strictly monotone (figure 11).

GDP time series. The Huber and L2 loss worked decently on these time series and produced an economically sensible segmentation: for instance, the models identified changepoints in the Germany GDP time in the beginning of the 2000's and at the end of the 2010's, which matches the periods of Germany's economic golden age and its industrial model crisis (see figure 8). However, the Biweight loss consistently resulted in undersegmentation. Besides, a drawback of using these methods on GDP is that since GDP is available at a low-frequency, and our methods require the intervals to be greater than a certain value of β/K, the detection of a changepoint takes time, as seen in figure 12. Hence, this method isn't suited for real-time detection of low-frequency series.

Finally, we tried to systematize the selection of parameters. First, we did it through a classical grid search on cost minimization on parameters β and K that were relatively close to the original parameters (maximum 10x the original beta), thinking it would reduce the number of changepoints. However these results proved very dissatisfying (figure 9), as the cross-validation would nearly always recover the minimal available beta - which is predictable since having a higher beta creates a larger cost. We therefore tried another method : an elbow method, detecting a stabilized number of changepoints between one changepoint and half the maximum numbers of changepoints used in the range of parameters. This method shrinks automatically the number of changepoints and gave economically sensible results for the biweight and huber loss (figure 10), although not as good as our by-hand manipulation.

4.3 Discussion

In summary, the method proposed in the article works to detect changepoints in the presence of outliers in economic data, but it has two major limitations. Firstly, the performance of the method will greatly depend on the data, and biweight loss does not seem to work particularly better than the others, contrary to the claims made in the article: satisfactory results are obtained on simulated data and GDP data with L2 and Huber, but disappointing results with biweight. Conversely, correct results are obtained on inflation data with Biweight and Huber, but not with L2. Secondly, these methods are very sensitive to the choice of parameters β and K. A small variation in these parameters will lead to over- or undersegmentation, and it is difficult to find the sweet spot that gives a meaningful segmentation: it can take time to iterate manually on the parameters, and it requires having a priori knowledge of the number of changepoints we want. If we have no prior knowledge of the desired number of change points, cross-validation or the elbow method can work, but we did not obtain outstanding results; it tended to over-segment and was very costly to run.

Other directions for future work should focus on designing a more efficient and less costly cross-validation method for choosing the parameters β and K. It would also be interesting to test other robust statistical losses than Huber and biweight, as this could potentially yield more satisfactory results.

References

[1] Paul Fearnhead and Guillem Rigaill. Changepoint detection in the presence of outliers. *Journal of the American Statistical Association*, 114(525):169–183, 2019.

[2] Piotr Fryzlewicz. Wild binary segmentation for multiple change-point detection. *The Annals of Statistics*, 42(6), 2014.

[3] Simon C. Smith, Allan Timmermann, and Stijn Van Nieuwerburgh. Break risk. *The Review of Financial Studies*, 34(4):2045–2100, 2021.

5 Appendix

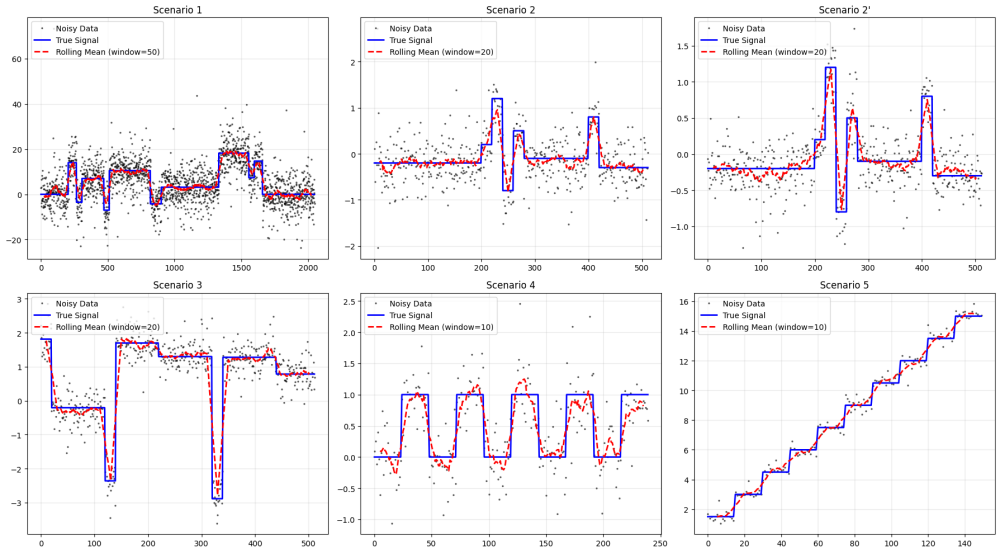

Figure 1: Simulated Scenarios used: data is not stationary

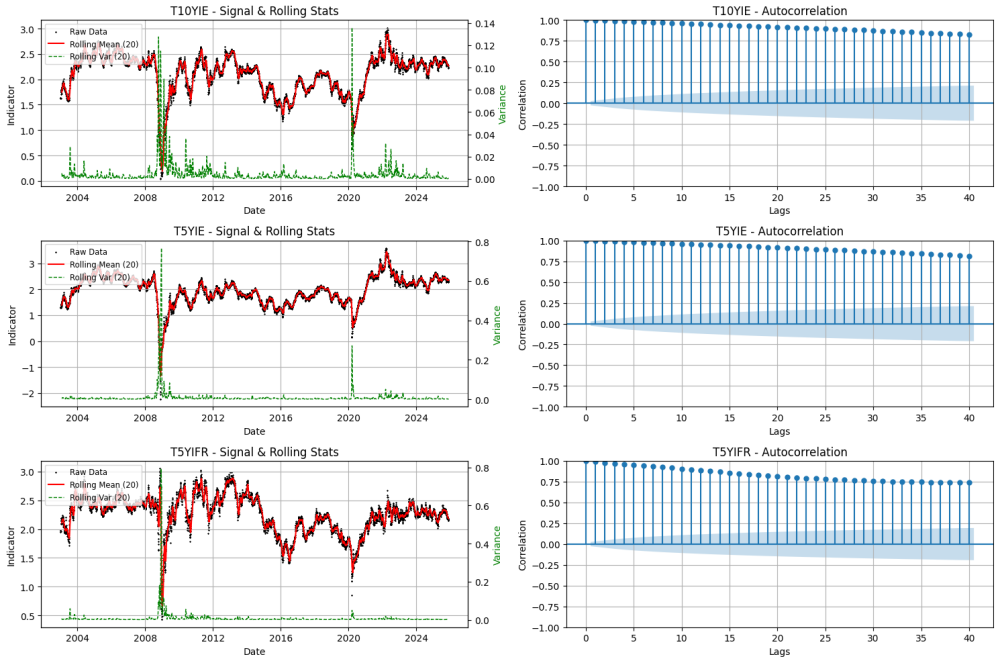

Figure 2: Inflation-related data used: data is not stationary

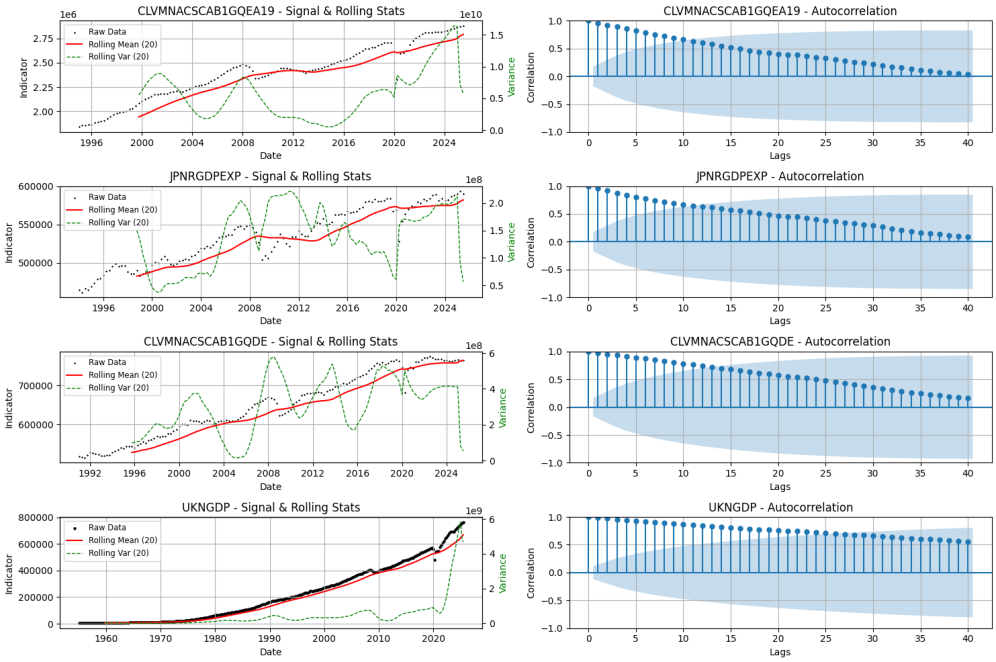

Figure 3: GDP-related data used: data is not stationary

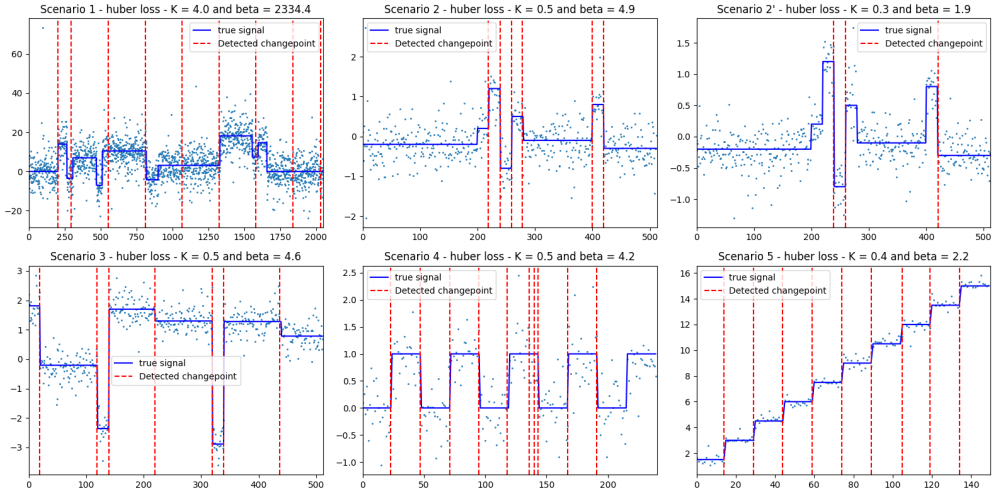

Figure 4: Results on synthetic data with the Huber loss after adjusting parameters iteratively

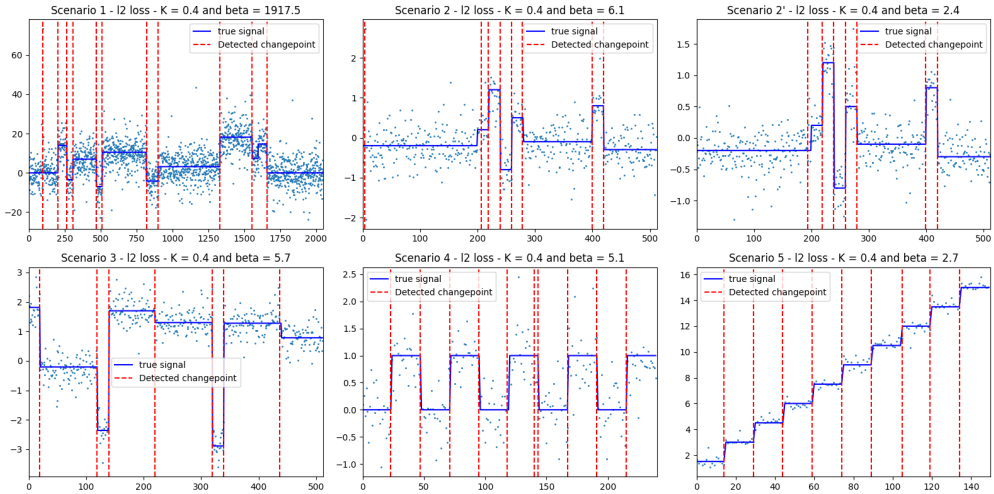

Figure 5: Results on synthetic data with the L2 loss after adjusting parameters iteratively

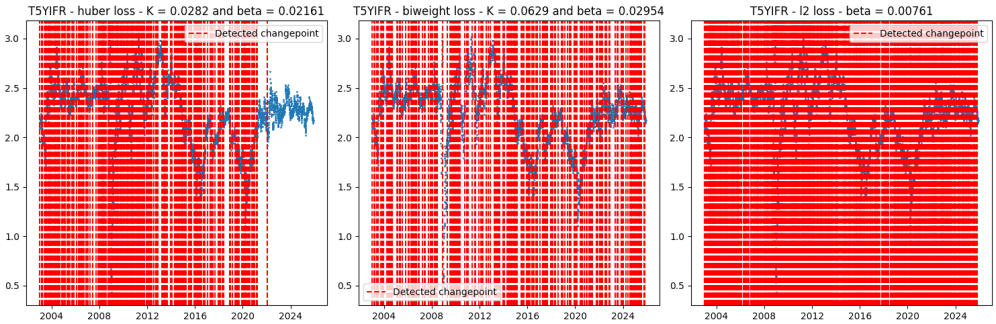

Figure 6: Over-segmentation problem: inflation expectations changepoint detection with parameters from the paper

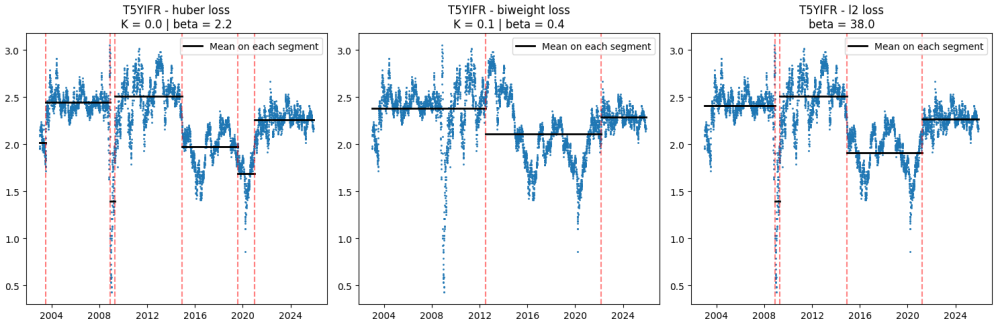

Figure 7: Inflation expectations changepoint detection with manually chosen parameters

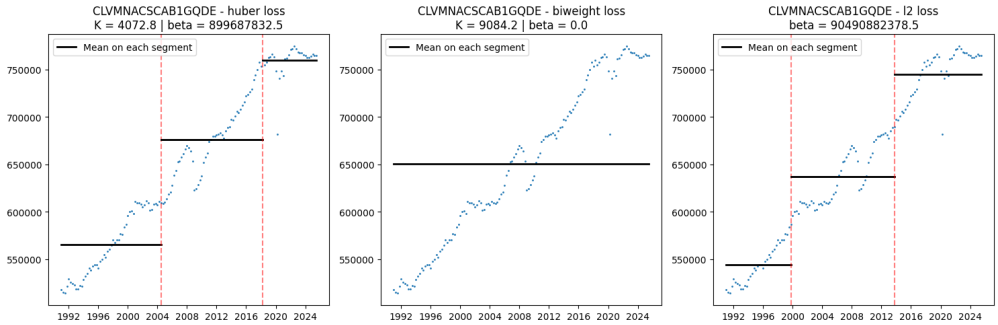

Figure 8: Germany GDP changepoint detection with manually chosen parameters: failure of biweight loss

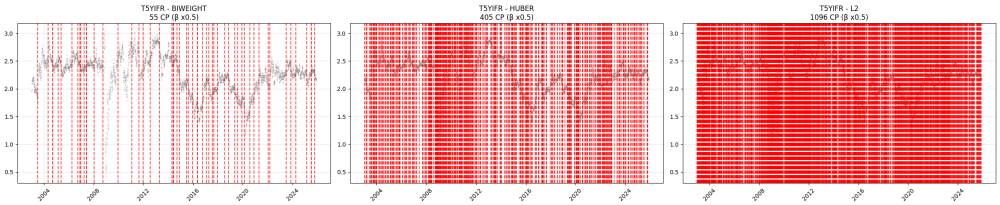

Figure 9: Over-segmentation problem: original cross-validation on inflation expectations

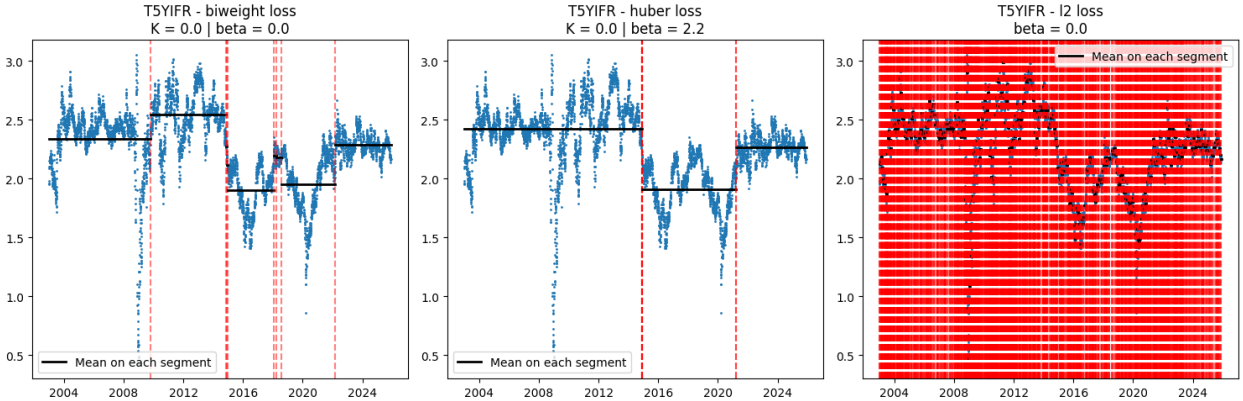

Figure 10: Elbow method results for inflation expectations

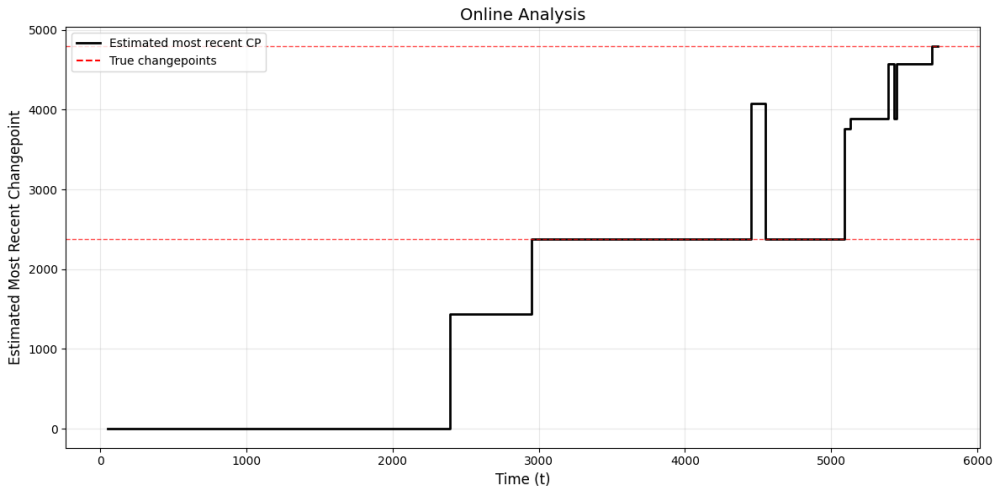

Figure 11: Inflation expectation : online analysis (Huber loss)

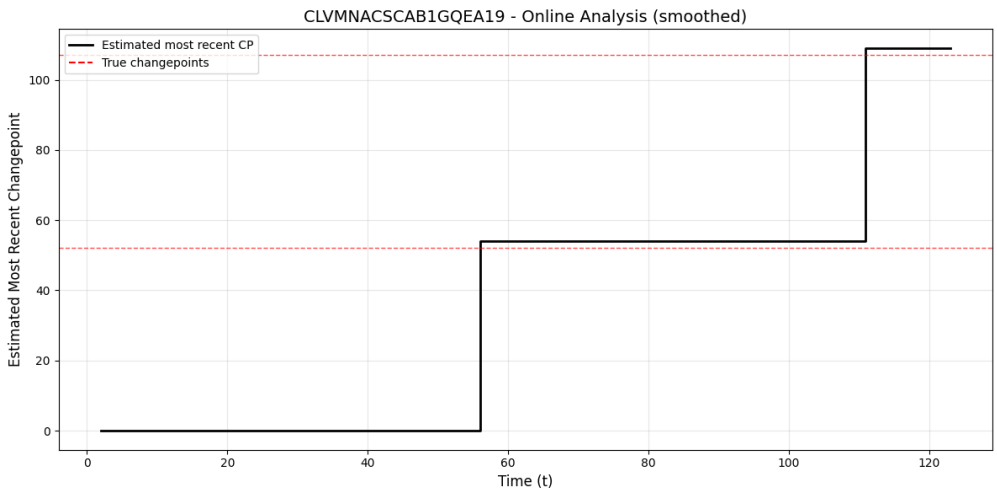

Figure 12: GDP : online analysis (Huber loss)